SOCIOL 723: Social Statistics II

Week 6 — Matching, Propensity Scores, and Weighting

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Today

Where We Are

Last week we established that under **conditional ignorability**, **positivity**, and **SUTVA**,

$$\text{ATE} = E\Big[E[Y_i \mid D_i = 1, X_i] - E[Y_i \mid D_i = 0, X_i] \Big].$$

Today we take that identification result as *given* and ask a purely practical question: how should we actually do the adjustment?

Four answers, and they are more alike than they look

1. **Matching**: find comparison units that look like the treated ones.
2. **Regression / g-computation**: model $E[Y \mid D, X]$ and average.
3. **Weighting**: reweight the sample so treatment is unrelated to X.
4. **Doubly robust**: do both, and get insurance.

All four estimate the *same* identified quantity. They differ in what they model, what weights they put on covariate cells, what they extrapolate, and how they fail.

Anatomy

Cell-by-Cell Comparison

Suppose X_i takes finitely many values and write the cell-level effect

$\delta(X_i = x) \equiv E[Y_i(1) - Y_i(0) \mid X_i = x] = E[Y_i \mid D_i = 1, X_i = x] - E[Y_i \mid D_i = 0, X_i = x]$,

where the second equality is conditional ignorability.

Every method today is a choice of weights on $\delta(X_i = x)$:

$$\delta_{\text{ATE}} = \sum_x \delta(X_i = x) \Pr(X_i = x)$$

$$\delta_{\text{ATT}} = \sum_x \delta(X_i = x) \Pr(X_i = x \mid D_i = 1)$$

The ATE weights cells by how common they are in the population; the ATT weights them by how common they are *among the treated*. Same ingredients, different recipe.

Anatomy of the Matching Estimand

Apply Bayes' rule to the ATT weights:

$$\Pr(X_i = x \mid D_i = 1) = \frac{\Pr(D_i = 1 \mid X_i = x) \Pr(X_i = x)}{\Pr(D_i = 1)}.$$

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Substituting,

$$\boxed{\delta_{\text{ATT}} = \frac{\sum_x \delta(X_i = x)\ \Pr(D_i = 1 \mid X_i = x)\ \Pr(X_i = x)}{\sum_x \Pr(D_i = 1 \mid X_i = x)\ \Pr(X_i = x)}}$$

The matching-based ATT weights each cell effect in proportion to the probability of treatment in that cell — that is, in proportion to the propensity score $e(x)$. Cells where treatment is common count for more, which is exactly what "the effect on the treated" should mean.

Anatomy of Regression Adjustment: Setup

Now do the same accounting for regression. To avoid conflating the weighting question with a functional-form question, specify a model *saturated in* X_i:

$$Y_i = \sum_x \mathbf{1}(X_i = x)\,\alpha_x \;+\; \delta_R\, D_i \;+\; \epsilon_i.$$

By Frisch–Waugh–Lovell (Week 1), with $\check{D}_i \equiv D_i - E[D_i \mid X_i]$ the residualized treatment,

$$\delta_R = \frac{\mathrm{Cov}(Y_i, \check{D}_i)}{\mathrm{Var}(\check{D}_i)} = \frac{E\big[\check{D}_i\, E[Y_i \mid D_i, X_i]\big]}{E\big[\check{D}_i^2\big]} = \frac{E\big[(D_i - E[D_i \mid X_i])\, E[Y_i \mid D_i, X_i]\big]}{E\big[(D_i - E[D_i \mid X_i])^2\big]}.$$

So the question becomes: what does the numerator simplify to?

Anatomy of Regression Adjustment: the Numerator

Because D_i is binary, the conditional expectation decomposes exactly:

$$\begin{aligned} E[Y_i \mid D_i, X_i] &= E[Y_i \mid D_i = 0, X_i] + D_i\Big(E[Y_i \mid D_i = 1, X_i] - E[Y_i \mid D_i = 0, X_i]\Big) \\ &= E[Y_i \mid D_i = 0, X_i] + D_i\,\delta(X_i). \end{aligned}$$

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Substitute into the numerator:

$$\begin{aligned} & E\Big[(D_i - E[D_i \mid X_i])\big(E[Y_i \mid D_i = 0, X_i] + D_i\delta(X_i)\big)\Big] \\ &= \underbrace{E\Big[(D_i - E[D_i \mid X_i])\,E[Y_i \mid D_i = 0, X_i]\Big]}_{=\,0 \text{ (a function of } X_i \text{ only)}} + E\Big[(D_i - E[D_i \mid X_i])\,D_i\,\delta(X_i)\Big] \\ &= E\Big[(D_i - E[D_i \mid X_i])^2\,\delta(X_i)\Big], \end{aligned}$$

where the last step uses $E[(D_i - E[D_i \mid X_i])D_i \mid X_i] = \mathrm{Var}(D_i \mid X_i)$.

Anatomy of Regression Adjustment: the Result

Putting numerator and denominator together and applying iterated expectations:

$$\delta_R = \frac{E\big[(D_i - E[D_i \mid X_i])^2\, \delta(X_i)\big]}{E\big[(D_i - E[D_i \mid X_i])^2\big]} = \boxed{\frac{E\big[\sigma_D^2(X_i)\, \delta(X_i)\big]}{E\big[\sigma_D^2(X_i)\big]}}$$

where $\sigma_D^2(x) \equiv \mathrm{Var}(D_i \mid X_i = x)$. Since D_i is binary, $\sigma_D^2(x) = e(x)(1 - e(x))$, so

$$\delta_R = \frac{\sum_x \delta(X_i = x)\, e(x)\big(1 - e(x)\big)\ \Pr(X_i = x)}{\sum_x e(x)\big(1 - e(x)\big)\ \Pr(X_i = x)}.$$

Regression weights each cell by the *conditional variance* of treatment. Compare with matching, which weighted by $e(x)$ itself. Cells with $e(x) \approx 0.5$ dominate the regression; cells that are nearly all treated or nearly all control contribute almost nothing — to either estimator.

Two Consequences Worth Stating Plainly

- **Regression and matching answer different questions** even under identical assumptions and with the same saturated specification. Neither is "the" ATE unless effects are homogeneous.
- **The regression weight $e(x)(1-e(x))$ vanishes at $e(x) \in \{0,1\}$**, so cells with no comparison units drop out of δ_R silently. Positivity failures do not produce an error message; they produce a quietly redefined estimand.
- Matching is different: a cell with $e(x) \approx 1$ carries a *large* ATT weight. But at $e(x) = 1$ there is no control unit, so $\delta(x)$ is simply not identified — the estimator has nothing to average, and the honest response is to say which cells you dropped.

With continuous covariates

The anatomy still holds, as an integral:

$$\delta_R = \frac{\int \delta(x)\, e(x)\big(1-e(x)\big) f(x)\, dx}{\int e(x)\big(1-e(x)\big) f(x)\, dx}.$$

Regression as Fuzzy Matching

With continuous X, exact common support at every $X_i = x$ is impossible in a finite sample. Positivity becomes the requirement that $0 < e(x) < 1$ on the support, so that in a *neighbourhood* of any covariate value both treated and untreated units occur:

$$E[Y_i(d) \mid X_i = x] = \lim_{\varepsilon \to 0} E\big[Y_i(d) \mid X_i \in (x - \varepsilon, x + \varepsilon)\big].$$

- In that sense, regression with continuous covariates *is* a fuzzy matching method, giving the largest weights to neighbourhoods with the most balanced overlap.
- Without common support, regression still returns a number — by extending the fitted control line into the treated region. That recovers the truth only if the control surface $E[Y \mid D = 0, X]$ is correctly specified out there *and* the treatment effect is constant across X_i — two functional-form claims, neither checkable where there are no data.
- This is the deepest reason to look at overlap before running anything.

Does It Matter Empirically? Angrist (1998)

Effect of *voluntary* military service on 1988–91 Social Security earnings, among men who applied to enter the armed forces 1979–82. Matching and regression both control for year of birth, education at application, and AFQT score.

	Mean earnings	Difference in means	Matching estimate	Regression estimate	Regression − matching
Whites	14,537	1,233.4	−197.2	−88.8	108.4
		(60.3)	(70.5)	(62.5)	(28.5)
Non-whites	11,664	2,449.1	839.7	1,074.4	234.7
		(47.4)	(62.7)	(50.7)	(32.5)

Adapted from Angrist (1998), Tables II and V. $n = 128{,}968$ whites, 175,262 non-whites.

- Adjustment reverses the sign for whites — selection into service is strongly positive on earnings potential.
- Matching and regression differ by about 100–235 dollars, and the *difference* is statistically significant. Small, but exactly the magnitude the weighting anatomy predicts.

Matching

The Idea, in Its Simplest Form

Suppose X_i takes finitely many values. For each cell x, compute the difference in means, then average the cell-level differences with weights of your choosing:

$$\hat{\tau}_{\text{ATE}} = \sum_x \hat{\Pr}(X_i = x)\,[\bar{Y}_{1x} - \bar{Y}_{0x}], \qquad \hat{\tau}_{\text{ATT}} = \sum_x \hat{\Pr}(X_i = x \mid D_i = 1)\,[\bar{Y}_{1x} - \bar{Y}_{0x}].$$

- This is exact matching, and it is the cleanest possible implementation of "compare like with like."
- Notice: the only difference between the ATE and the ATT is the weights on the cells. Choose them deliberately.
- Recall from Week 1 that OLS with a saturated X reports its own variance-weighted average of the same cell effects. Matching lets you take the weights back.
- The catch: with more than a handful of covariates, most cells are empty.

The Curse of Dimensionality

With p binary covariates there are 2^p cells: $p = 10$ gives 1,024, $p = 20$ gives over a million. Exact matching becomes impossible long before the covariates run out.

Practical responses, in rough order of desperation:

- **Coarsened exact matching** (CEM): bin continuous covariates coarsely — age in five-year bands, income in quintiles — then match exactly on the bins. Transparent, and you see immediately what you had to discard.
- **Nearest-neighbour matching** on a distance: Euclidean, or *Mahalanobis* $\left[(X_i - X_j)'\hat{\boldsymbol{\Sigma}}^{-1}(X_i - X_j)\right]^{1/2}$, which scales by the covariance so no variable dominates by its units.
- **Propensity score matching**: collapse X to a single number — the next section.

Every one of these is a way of admitting that positivity fails somewhere. Being explicit about where is a feature.

Choices That Actually Matter

- **With or without replacement.** With replacement gives better matches and lower bias but higher variance, and the same control unit may be reused many times (which the standard errors must account for).
- **How many matches per treated unit.** More matches reduce variance and increase bias.
- **Calipers.** Refuse matches beyond some distance. This improves comparability but changes the estimand: you are now describing whatever subpopulation survived.
- **Bias correction.** Nearest-neighbour matching on continuous covariates has a bias of order $n^{-1/p}$ that does not vanish fast enough; regress the residual difference on the covariate gap to remove it.

Report what matching discarded

The number of treated units with no acceptable match is a substantive result. It tells your reader exactly where the data cannot speak.

Matching Is Preprocessing, Not Estimation

Ho, Imai, King, and Stuart (2007) make the point that should organize your workflow: matching is a *nonparametric preprocessing step* that reduces how much your final model has to do.

1. Match, using no outcome data at all. This is the design stage; it can be iterated freely without p-hacking, because you cannot see the answer.
2. Check **balance**: are covariate distributions similar in the matched treated and control groups? Standardized mean differences below 0.1, variance ratios near 1, and eyeballed density overlaps.
3. *Then* run your regression on the matched sample. Any remaining imbalance is handled by the model, and because imbalance is now small, the answer is insensitive to how you specify it.

Never use a balance t-test. Balance is a property of the sample you have, not a hypothesis about a population; its p-value falls mechanically as matching discards observations (Imai, King, and Stuart 2008).

Propensity Score

A Remarkable Theorem

Define the propensity score

$$e(x) \equiv \Pr(D_i = 1 \mid X_i = x).$$

Rosenbaum and Rubin (1983)

If $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i \mid X_i$, then

$$(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i \mid e(X_i).$$

- Conditioning on a *single number* suffices. A p-dimensional matching problem collapses to a one-dimensional one.
- $e(X)$ is a *balancing score*: within strata of $e(X)$, the whole covariate vector is distributed identically in the two groups.
- This is why the propensity score is everywhere. It is also why it is routinely oversold — it does nothing about *unobserved* confounders.

Proof of the Balancing Property ★

It suffices to show $\Pr(D_i = 1 \mid X_i, e(X_i)) = \Pr(D_i = 1 \mid e(X_i))$; the left side is $e(X_i)$ by definition. For the right side, use iterated expectations:

$$\Pr\left(D_i = 1 \mid e(X_i)\right) = E\left[\, E[D_i \mid X_i] \mid e(X_i)\right] = E\left[\, e(X_i) \mid e(X_i)\right] = e(X_i).$$

So $D_i \perp\!\!\!\perp X_i \mid e(X_i)$: within a stratum of the propensity score, treatment carries no further information about X.

Combining with conditional ignorability on X gives ignorability on $e(X)$. ■

- Note this is a statement about the *true* $e(x)$. In practice we estimate it, and the properties of $\hat{e}$ matter — Week 13.
- Any function of X that is finer than $e(X)$ is also a balancing score; $e(X)$ is the coarsest one.

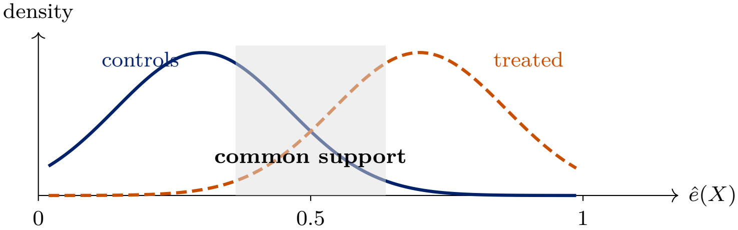
Estimating and Checking It

- Usually a logit of D on X (Week 4). The model is not judged by fit. A propensity model with a high AUC is often a bad sign — it means treated and control units are easy to tell apart, i.e. poor overlap.
- Judge it by **balance**: does weighting or matching on $\hat{e}$ equalize the covariate distributions? Iterate on the specification until it does, adding interactions and polynomials as needed.
- Plot the two distributions of $\hat{e}$. Mass near 0 among the treated, or near 1 among the controls, is a positivity failure staring at you.

The propensity score paradox

The better your propensity model predicts treatment, the less common support you have, and the less the data can tell you. Prediction quality and identification pull in opposite directions.

Overlap: What to Do When It Fails



- **Trim** to a range of $\hat{e}$ (say $[0.1, 0.9]$), and say so. You have changed the estimand to that subpopulation.
- **Redefine the population** to the region where comparison is possible — often the honest and more interesting move.
- **Change the estimand**: the ATT needs overlap only for the treated; the "overlap-weighted" estimand (Li, Morgan, and Zaslavsky 2018) deliberately targets the region where comparison is best.

Weighting

Inverse Probability Weighting

Rare treated units in a covariate cell should count for more, because they stand in for everyone like them who was not treated. Weight each unit by the inverse of the probability of the treatment it actually received:

$$\boxed{\hat{\tau}_{\text{IPW}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{D_i Y_i}{\hat{e}(X_i)} - \frac{(1-D_i)Y_i}{1-\hat{e}(X_i)}\right]}$$

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Why it works. For the treated arm, condition on X_i first:

$$E\left[\frac{D_i Y_i}{e(X_i)}\right] = E\left[\frac{E[D_i \mid X_i]\, E[Y_i(1) \mid X_i]}{e(X_i)}\right] = E\big[E[Y_i(1) \mid X_i]\big] = E[Y_i(1)],$$

where ignorability lets us factor D_i apart from $Y_i(1)$ given X_i, and consistency replaces Y_i by $Y_i(1)$ among the treated. Same for the control arm. ■

This is the Horvitz–Thompson estimator from survey sampling, applied to a “sample” created by the treatment assignment mechanism.

One Class of Estimands, Three Members

Hirano, Imbens, and Ridder (2003): IPW for the ATE, IPW for the ATT, and the regression estimand are all members of one weighted-average family,

$$E\left[g(X_i)\left(\frac{Y_i D_i}{e(X_i)} - \frac{Y_i(1-D_i)}{1-e(X_i)}\right)\right],$$

differing only in the choice of $g(\cdot)$:

Estimand	$g(X_i)$
ATE	1
ATT	$\frac{e(X_i)}{\Pr(D_i=1)}$
Regression	$\frac{e(X_i)(1-e(X_i))}{E\left[e(X_i)(1-e(X_i))\right]}$

This is the same anatomy result again, now written as a weighting function. Regression is not an alternative to weighting; it is a particular weighting you did not choose.

Stabilized IPW

The raw weights $1/\hat{e}$ explode as $\hat{e} \to 0$: one observation with $\hat{e} = 0.001$ carries the weight of a thousand. Multiply by the marginal treatment probabilities:

$$w_i^{\text{SIPW}} = \begin{cases} \dfrac{\Pr(D_i = 1)}{e(X_i)}, & D_i = 1 \\ \dfrac{\Pr(D_i = 0)}{1 - e(X_i)}, & D_i = 0. \end{cases}$$

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The rescaling cancels in the ratio, so the estimand is unchanged:

$$\frac{E\left[\frac{\Pr(D_i=1)}{e(X_i)} D_i Y_i\right]}{E\left[\frac{\Pr(D_i=1)}{e(X_i)} D_i\right]} = \frac{E\left[\frac{1}{e(X_i)} D_i Y_i\right]}{E\left[E\left[\frac{1}{e(X_i)} D_i \,\middle|\, X_i\right]\right]} = E\left[\frac{D_i Y_i}{e(X_i)}\right] = E[Y_i(1)],$$

because the denominator is $E[e(X_i)/e(X_i)] = 1$.

Horvitz–Thompson versus Hájek

The variance gain comes from *normalizing*, not from the constant. Compare a raw average with a ratio:

$$\hat{\delta}_{\text{SIPW}} = \frac{\sum_i w_i^{\text{SIPW}} D_i Y_i}{\sum_i w_i^{\text{SIPW}} D_i} - \frac{\sum_i w_i^{\text{SIPW}} (1 - D_i) Y_i}{\sum_i w_i^{\text{SIPW}} (1 - D_i)}.$$

- Note the arithmetic: the constant $\Pr(D_i = 1)$ appears in both numerator and denominator of the ratio and *cancels*. So for this estimator, "stabilizing" changes nothing; what matters is that we divided by $\sum_i w_i$ rather than by n.
- That normalization is the real gain. The Hájek estimator's weights sum to the sample size in each arm, so the weighted pseudo-population is not artificially inflated, the estimate respects the range of Y, and its variance is typically much smaller.
- Stabilized weights *do* matter when the weights enter a regression rather than a ratio — the marginal structural model setting.
- Neither fixes poor overlap. Extreme weights survive both; they are a positivity problem wearing a disguise. Diagnose with the largest weights and the *effective sample size* $\left(\sum_i w_i\right)^2 / \sum_i w_i^2$.

IPW Is a Weighted Regression

Define the Horvitz–Thompson transformation

$$H_i = \frac{D_i}{e(X_i)} - \frac{1 - D_i}{1 - e(X_i)}.$$

Then $E[H_i Y_i] = E[Y_i(1) - Y_i(0)]$, which is the IPW estimand written in one term. Equivalently:

The coefficient on D_i from a **weighted least squares** regression of Y_i on D_i (with an intercept) and weights $1/\hat{e}(X_i)$ for the treated, $1/(1 - \hat{e}(X_i))$ for the controls, equals the *normalized* (Hájek) IPW estimator — a difference in weighted means. It is not the raw Horvitz–Thompson average, which divides by n instead.

- Practically convenient: you can run IPW with `lm(y ~ d, weights = w)` and add covariates for extra protection.
- Conceptually important: it makes the continuity between "weighting" and "regression" explicit, and it is the bridge to the doubly robust estimator below.
- Note the standard errors from `lm` will be wrong — they ignore that $\hat{e}$ was estimated. Bootstrap the whole pipeline.

Weighting to Different Estimands

Every adjustment method is a choice of weights on covariate cells. Writing them side by side makes the choice explicit:

Estimand	Weight on treated	Weight on controls
ATE	$1/e(X)$	$1/(1-e(X))$
ATT	1	$e(X)/(1-e(X))$
ATU	$(1-e(X))/e(X)$	1
Overlap (ATO)	$1-e(X)$	$e(X)$

- The ATT weights leave the treated alone and reweight controls to look like them — which is what “the effect on the treated” means.
- The *overlap* weights are bounded and largest at $e = 0.5$; they never explode. Note that they reproduce exactly the regression weighting $e(x)(1-e(x))$ from the anatomy — so “ATO” is the estimand OLS was targeting all along, now chosen deliberately (Li, Morgan, and Zaslavsky 2018).

G-Computation: the Other Half

Instead of modelling treatment, model the outcome. Under ignorability and common support, fit

$$m_d(x) = E[Y_i \mid D_i = d, X_i = x], \qquad d \in \{0,1\},$$

then *impute both potential outcomes for everyone* and average:

$$\hat{\tau}_{\text{g-comp}} = \frac{1}{n}\sum_{i=1}^{n}\left[\hat{m}_1(X_i) - \hat{m}_0(X_i)\right].$$

- This is exactly the "observed-value" predicted-probability procedure from Week 4, and in the linear case it *is* regression adjustment.
- It uses the whole sample and is efficient when the outcome model is right. It extrapolates silently when it is not: $\hat{m}_1(x)$ is computed even for x where nobody was treated.
- Note the symmetry: IPW models $\Pr(D \mid X)$ and never touches Y; g-computation models $E[Y \mid D, X]$ and never touches assignment. Each is exposed to a different misspecification — which is the opening for the next section.

Doubly Robust

Why Not Both?

IPW is consistent only if $e(X_i)$ is right. G-computation is consistent only if $m_d(X_i)$ is right. Combine them so that *either* suffices.

Given the two nuisance functions $m_d(x) = E[Y_i \mid D_i = d, X_i = x]$ and $e(x) = \Pr(D_i = 1 \mid X_i = x)$, the augmented IPW estimand is

$$\theta = E\Bigg[\underbrace{m_1(X_i) - m_0(X_i)}_{\text{g-computation}} + \underbrace{\frac{D_i(Y_i - m_1(X_i))}{e(X_i)} - \frac{(1-D_i)(Y_i - m_0(X_i))}{1 - e(X_i)}}_{\text{IPW-weighted residuals}}\Bigg]$$

Double robustness

$\theta = E[Y_i(1) - Y_i(0)]$ if either m_d or e is correctly specified — not necessarily both. And when both are right, AIPW attains the semiparametric efficiency bound.

Case 1: the Outcome Model Is Correct

Suppose m_1 is right but e is arbitrary. Look at the augmentation term for the treated arm, conditioning on X_i:

$$E\left[\frac{D_i}{e(X_i)}(Y_i - m_1(X_i)) \;\middle|\; X_i\right] = \frac{1}{e(X_i)} \Pr(D_i = 1 \mid X_i)\, E[Y_i - m_1(X_i) \mid D_i = 1, X_i]$$

$$= \frac{1}{e(X_i)} E[D_i \mid X_i] \cdot \underbrace{0}_{\text{by definition of } m_1} = 0.$$

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$$= \frac{1}{e(X_i)} E[D_i \mid X_i] \cdot \underbrace{0}_{\text{by definition of } m_1} = 0.$$

The same argument gives $E\big[\frac{1-D_i}{1-e(X_i)}(Y_i - m_0(X_i)) \mid X_i\big] = 0$. Hence

$$\theta = E\big[m_1(X_i) - m_0(X_i)\big] = E[Y_i(1) - Y_i(0)].$$

The whole augmentation term vanishes, whatever e is. Note that this did not use a correct propensity score anywhere.

Case 2: the Propensity Score Is Correct

Now suppose e is right but m_1 is some wrong function. Then

$$E\left[\frac{D_i}{e(X_i)}(Y_i - m_1(X_i)) \;\middle|\; X_i\right] = \frac{1}{e(X_i)} E[D_i Y_i \mid X_i] - m_1(X_i).$$

Since $E[D_i Y_i \mid X_i] = e(X_i)\, E[Y_i(1) \mid X_i]$ by ignorability, this is

$$E[Y_i(1) \mid X_i] - m_1(X_i).$$

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$$E[Y_i(1) \mid X_i] - m_1(X_i).$$

Symmetrically the control term contributes $-\big(E[Y_i(0) \mid X_i] - m_0(X_i)\big)$. Adding everything up, the wrong m_1, m_0 cancel exactly:

$$\theta = E\Big[E[Y_i(1) \mid X_i] - E[Y_i(0) \mid X_i]\Big] = E[Y_i(1) - Y_i(0)]. \qquad \blacksquare$$

Two independent routes to the same answer. That is the insurance.

Why the Cancellation Matters Beyond This Week

Look again at what happened in the two cases. A first-order error in *one* nuisance function was wiped out by the other term. That is not a coincidence of AIPW; it is a property the estimating equation was built to have.

The consequence

The bias of AIPW is roughly the *product* of the two nuisance errors, not their sum. So if both $\hat{e}$ and $\hat{m}_d$ are merely *approximately* right, the bias is second-order small — much smaller than either model's own error.

- This is why doubly robust estimators are the default in modern practice even when nobody believes either model exactly.
- It is also what eventually licenses estimating $\hat{e}$ and $\hat{m}_d$ with flexible, machine-learning methods rather than logit and OLS. That is a Week 13 topic; the property has a name (*Neyman orthogonality*) and we will not need it before then.

An Honest Warning

Double robustness is often oversold.

- "Either model correct" is a low bar in theory and an unreachable one in practice: both models are almost certainly wrong, and then AIPW is inconsistent too.
- What it really buys is second-order bias: the bias is the *product* of the two models' errors, so it is small when both are approximately right. That is genuinely valuable, and it is the honest version of the claim.
- With extreme weights, AIPW can be more unstable than either component alone (Kang and Schafer 2007).
- None of this addresses unobserved confounding. Everything today assumes conditional ignorability, and no diagnostic can test it.

Sensitivity Analysis: Taking the Assumption Seriously

Since ignorability is untestable, ask instead: *how strong would an unobserved confounder have to be to overturn my conclusion?*

- **Rosenbaum bounds**: how much would two matched units have to differ in their odds of treatment before the inference flips?
- **Partial R^2 / robustness value** (Cinelli and Hazlett 2020): express the required confounder strength as the share of residual variation in D and in Y it would have to explain, then benchmark it against an *observed* covariate — "the confounder would need to be three times as strong as parental education."
- **E-value**: the minimum risk ratio a confounder would need with both treatment and outcome.

Report one. It converts an untestable assumption into a quantitative claim your reader can argue with.

Choosing Among the Methods

Method	Models	Main weakness
Exact / CEM matching	nothing	discards data; infeasible with many covariates
NN / Mahalanobis matching	a distance	bias with continuous X; variance from reuse
PS matching	$\Pr(D \mid X)$	sensitive to the PS model; discards controls
IPW	$\Pr(D \mid X)$	extreme weights when overlap is poor
Regression / g-computation	$E[Y \mid D, X]$	silent extrapolation; own weighting
AIPW	both	unstable weights; still needs ignorability

- If the answers differ a lot, that is worth chasing down. By the anatomy results, systematic differences can mean the cell-level effects $\delta(x)$ are heterogeneous and the methods weight them differently — but they can equally reflect a misspecified model, poor overlap, or sampling noise. Distinguish these before concluding anything.
- Report the primary estimator you pre-committed to, plus the others as robustness — never the reverse.

A Workflow

1. State the estimand (ATE? ATT? ATO? on which population?) before touching data.
2. Draw the DAG. Decide what to condition on — and what *not* to.
3. Estimate the propensity score. Look at overlap. Trim or redefine if needed and say so.
4. Match or weight. Check balance with standardized mean differences and density plots, never with t-tests.
5. Estimate with a doubly robust estimator; report matching, weighting, and regression alongside.
6. Get standard errors that respect the design: bootstrap the whole estimation pipeline for weighting and regression estimators, and cluster where treatment was assigned. Do not bootstrap nearest-neighbour matching — the ordinary bootstrap is invalid for it (Abadie and Imbens 2008); use their analytic variance instead.
7. Report a sensitivity analysis.

Looking Ahead

- Everything today rested on conditional ignorability: the claim that we measured every confounder. Matching, weighting, and AIPW make that assumption easier to implement — they do not make it more likely to be true.
- The anatomy results are the transferable lesson: *every* adjustment method is a weighting of cell-level effects, and knowing the weights tells you what your number means.
- **Weeks 7–11** abandon ignorability and look for special structure — an instrument, a discontinuity, parallel trends — that identifies an effect *despite* unobserved confounding.
- **Week 13** returns to today's AIPW estimator and asks what changes when the covariate set is large enough that logit and OLS will not do.

Problem Set 3

Assigned today, due Monday October 12. Covers Weeks 5–6.

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